AUTOSAR High-Level Design Specification: Engine Control Unit (ECU)
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1. Executive Overview
This document specifies the software architecture for the Engine Control Unit (ECU). The ECU manages fuel inje

2. Software Components
The architecture consists of the following primary software components:
- Component: EngineManagementSwc
- Component: FuelInjectionSwc
- Component: IgnitionControlSwc
- Component: ThrottlePositionSwc

3. Interfaces and Ports
- Interface: If_EngineSpeed (Provides engine RPM calculation signals to FuelInjectionSwc and IgnitionControlSw
- Interface: If_ThrottleCmd (Port: PPort_ThrottlePosition to RPort_ThrottleActuator).
- Interface: If_FuelInjectionTiming (Port: PPort_FuelPulseWidth).

4. Signal Specifications
- Signal: Sig_EngineRPM (Data type: uint16, Range: 0 - 8000 RPM)
- Signal: Sig_ThrottleAngle (Data type: float32, Range: 0.0 - 100.0 %)
- Signal: Sig_InjectionDuration (Data type: uint32, Unit: microseconds)

5. Safety & Fault Handling
- Functional Safety Target: ISO 26262 ASIL-C requirement for overspeed prevention.
- Fault Reaction: If Sig_EngineRPM exceeds 7500 RPM for more than 50ms, FuelInjectionSwc must enter limp-h

6. Dependencies & Data Flow
- Component FuelInjectionSwc depends on If_EngineSpeed and Sig_ThrottleAngle to compute Sig_InjectionDur
- Component IgnitionControlSwc relies on If_EngineSpeed to calculate ignition spark angle.